

Charles Ciampa

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Education

Northeastern University; Khoury College of Computer Sciences

Boston, MA

M.S. in Artificial Intelligence, Concentration in Machine Learning; GPA: 4.0

Expected December 2026

B.S. in Computer Science and Physics, Minor in Mathematics; Dean's List

August 2025

Related Courses: Autonomous Field Robotics, Robotic Science & Systems, Robotic Sensing & Navigation, Reinforcement Learning
Deep Learning, Computer Vision, Machine Learning & Data Mining, Natural Language Processing, Stochastic Processes, Electronics

Skills

Computer Languages: Python | Swift | Java | JavaScript | Rust | Typescript | LaTeX | HTML | CSS | Racket | R

Software: Git/GitHub | Claude Code | NumPy & Pandas | PyTorch | ROS2 | OpenCV | robosuite (MuJoCo) | React | Jira | Selenium

Experience

S1 Industries PBC Inc.

August 2024 – December 2024

Engineering Intern

Medford, MA

- Performed 10+ explorations of technologies and problems using methodology developed by MIT professor Luis Perez-Breva for his startups focused iTeams class, including reading research for each new technology and proposing potential companies to create
- Spearheaded the development of an Augmented Reality Tic-Tac-Toe game with options for multiplayer remote playing using SharePlay (with/without Apple Personas) for VisionOS's mixed immersive environment using Swift
- Created app specifications, testing procedures, and utilized TestFlight for internal testing with a small community of users to ensure the Tic-Tac-Toe app met requirements
- Launched an internal AI project using local LLMs to read technical documents and access the internet through a local instance of SearXNG and mapped over 70 separate AI and LLM tools and software with 3 views, including relevant logos, summaries, use cases, rankings, and applicable licenses related to usage or development

Genuity

July 2023 – December 2023

Medical Device Testing Engineer Associate

Sudbury, MA

- Engineered a well-documented Python program for point tracking in high-speed camera footage, offering users flexibility with customizable algorithms and seamless data export in multiple formats
- Developed a user-friendly interface and adaptable tracking settings, providing accessibility through Python scripts, command line interface, options for selection of specific video frame areas and time intervals for efficient analysis, and a graphical user interface (GUI), optimizing user interaction and usability
- Conducted rigorous testing on both software and catheter integration with the OCT device, ensuring robust verification and validation and meticulous data collection for quality assurance purposes
- Executed tests and collected data to secure FDA, BSI, and PMDA submissions for a new catheter

AI Club

May 2023 – April 2025

Founding Member and Vice President

Boston, MA

- Pioneered a project proposal, assignment, and mentoring system enabling club members to collaborate in teams and raise funding for AI projects, including SwarmScape (swarm robotics for greenhouse monitoring)
- Launched InnovAte, now an annual 2-day AI hackathon, attracting 80+ participants and 11 expert speakers/judges from industry and academia in its inaugural year
- Managed 8-person exec board to coordinate weekly meetings, schedule student events, organize lab tours, and plan future semesters

Projects

Language-Conditioned Diffusion Policy (charles.ciampa.me/projects/imitation-learning-diffusion)

March 2026 – April 2026

- Reimplemented a diffusion policy (PyTorch) with FiLM language conditioning to drive a 7-DoF Franka robotic arm on the 10 long-horizon tasks of LIBERO-10 (robosuite/MuJoCo), generating target poses and gripper states from vision and language
- Built a modular pipeline for swappable vision encoders; ResNet-18 from scratch reached 32.5% task success vs 0% for frozen SigLIP2
- Generated ~40 LLM paraphrases per task for a 4x gain in task success (32.5% vs 7.5%) over task-ID memorization

Estimating Trajectory From IMU (charles.ciampa.me/projects/imu)

September 2025 – December 2025

- Estimated vehicle trajectory from IMU data using a weighted-average complementary filter ($\alpha=0.98$) to fuse magnetometer and gyroscope headings, achieving position accuracy within 2 m of GPS for up to 6.8 seconds of dead reckoning
- Calibrated magnetometer via least-squares for hard/soft iron correction, validated on 18-minute dataset with 10+ turns
- Developed Python ROS2 Jazzy drivers for VN-100 IMU/GPS, implementing custom messages, NMEA parsing, UTM conversion
- Documented methodology and analysis in 7-page paper formatted to IEEE standards using LaTeX ([paper](#))

Robotic Photographer for 3D Gaussian Splatting (charles.ciampa.me/projects/robot_photographer)

March 2025 – April 2025

- Deployed ROS Noetic on a LoCoBot mobile robot (Create 3 base) to autonomously detect, approach, and circumnavigate objects via segmentation (no April tags), capturing RGBD imagery for 3D Gaussian Splatting reconstruction
- Implemented ZeroMQ networking to stream RGBD data to a remote workstation for a sub-minute capture workflow (team of 4)

Interests

Languages: English | Spanish (Fluent and Literate)

Other: Traveling | Cooking | Astronomy | Reading (Manga, Non-Fiction) | Open-World Video Games (Minecraft, Red Dead 2) | Rowing